# LESSON PLAN

## U/W WEAPON (38)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 09 |
| **Total No. of Lesson Plans** | 9 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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| **1. Topic** | Torque Tightening Procedure – Introduction to torque giving tools, Torque Calculation and Torque tightening practice. |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Torque Tightening Procedure – Introduction to torque giving tools, Torque Calculation and Torque tightening practice. |
| **5. Specific Objectives** | 38.1.1.1 Trainee will be able to explain principles of Correct Aircraft Maintenance Practices, tools and their usage |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Torque tightening practice on simulated fasteners using calibrated torque wrenches; Safety precautions include PPE usage and zeroing tools; Instructor supervision ensures correct technique and reading interpretation. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What is the primary purpose of torque tightening in aircraft maintenance?
(b) Name two types of torque giving tools used in aviation.
(c) Why is it critical to follow correct aircraft maintenance practices regarding fasteners?

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Introduction to Torque Giving Tools<br>• Selection and Calibration of Torque Wrenches<br>• Principles of Correct Aircraft Maintenance Practices | L & Ds | PPT, IB | 15 min |
| (b) | • Torque Calculation Formulas<br>• Conversion of Units and Factors<br>• Interpretation of Torque Specifications | L & Ds | PPT, PDF | 20 min |
| (c) | • Hands-on Torque Tightening Practice<br>• Verification of Applied Torque Values<br>• Common Errors and Correction Techniques | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the relationship between torque, force, and lever arm length.
(b) Describe the procedure for verifying that a fastener has been tightened to the correct specification.
(c) Identify three common errors that can occur during torque tightening and their consequences.

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| **12. Home Assignment and Follow Up** | Prepare a brief summary of the torque calculation method discussed and outline the key steps for the next lesson plan (LP 02) regarding specific aircraft fastener applications. |
| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | End of subject | Refer to the master syllabus reference list | - |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min